NETWORK SECURITY LAB PROJECT

VLAN Segmentation, DHCP, ACL & Attack Simulation

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1. INTRODUCTION

This project demonstrates the design and implementation of a secure network using VLAN segmentation, DHCP, inter-VLAN routing, and access control mechanisms. It also includes simulation of network attacks and implementation of defensive techniques.

2. OBJECTIVES

- To design a segmented network using VLANs
- To configure DHCP for dynamic IP allocation
- To implement inter-VLAN routing using router-on-a-stick
- To apply ACLs for access control
- To simulate network attacks
- To implement port security for unauthorized access prevention

3. TOOLS & TECHNOLOGIES

- **Cisco Packet Tracer**: A simulation tool used to design, configure, and troubleshoot virtual networks.
- **Virtual LAN (VLAN)**: A method of dividing a physical network into separate logical networks for better security and performance.
- **Dynamic Host Configuration Protocol (DHCP)**: A protocol that automatically assigns IP addresses and network settings to devices on a network.
- **Access Control List (ACL)**: A set of rules used to control and filter incoming and outgoing network traffic.

4. NETWORK DESIGN

- 4 VLANs:
  - VLAN 10 → HR
  - VLAN 20 → IT
  - VLAN 30 → Admin
  - VLAN 40 → Server
- Router used for inter-VLAN routing
- Switch used for VLAN segmentation

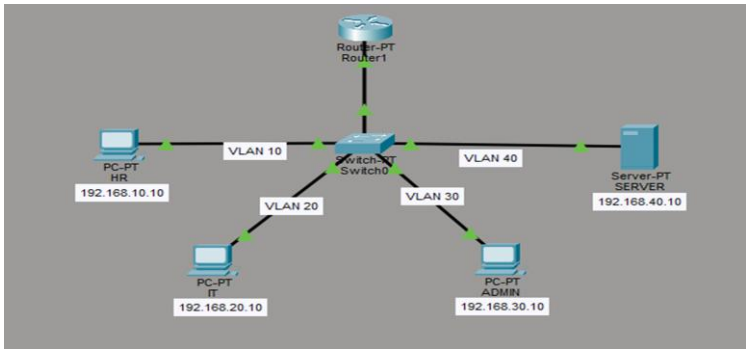


Figure 1: Network Design using cisco Packet Tracer

5. IMPLEMENTATION

VLAN Creation

- First, four **Virtual LAN (VLAN)**s (10, 20, 30, 40) were created on the switch to represent different departments (HR, IT, Admin, Server).
- Then, specific switch ports were assigned to each VLAN so that devices connected to those ports belong to their respective department's network.

Trunk Configuration

- A trunk link was configured between the switch and the router, allowing multiple VLANs to pass through a single physical connection.
- The trunk port was set to allow traffic from VLANs 10, 20, 30, and 40, enabling the router to receive and route data between all VLANs (inter-VLAN communication).

```
Switch>enable
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------
1    default                          active    Fa4/1, Fa5/1
10   HR                               active    Fa1/1
20   IT                               active    Fa2/1
30   ADMIN                            active    Fa3/1
40   SERVER                           active    Eth6/1
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

Figure 2: VLAN Configuration

```
Switch#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       1-1005

Port        Vlans allowed and active in management domain
Fa0/1       1,10,20,30,40

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       1,10,20,30,40
```

Figure 3: Trunk Information

6. DHCP Configuration

- Separate DHCP pools were created for each VLAN using **Dynamic Host Configuration Protocol (DHCP)**, ensuring that each department (VLAN 10, 20, 30, 40) receives IP addresses from its own network range.
- A default gateway was assigned in each DHCP pool, which points to the router interface responsible for that VLAN, allowing devices to communicate outside their VLAN.
- With automatic IP allocation, devices connected to the network are dynamically assigned IP addresses, subnet masks, and gateway settings without manual configuration.

```
Router>enable
Router#show running-config | include dhcp
ip dhcp excluded-address 192.168.10.1
ip dhcp excluded-address 192.168.20.1
ip dhcp excluded-address 192.168.30.1
ip dhcp pool HR
ip dhcp pool IT
ip dhcp pool ADMIN
```

Figure 4: DHCP Configuration

```
Router#show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
192.168.10.2     0090.2B38.1E6D           --                     Automatic
192.168.20.2     00D0.FF83.6BB2           --                     Automatic
192.168.30.2     0004.9A98.E5A0           --                     Automatic
Router#
```

Figure 5: DHCP Binding

7. Inter-VLAN Routing

- The **router-on-a-stick** method was used, where a single physical router interface handles traffic for multiple Virtual LAN (VLAN)s by using a trunk link to the switch.
- Sub-interfaces were created on the router—one for each VLAN (10, 20, 30, 40)—and each sub-interface is configured with a VLAN ID using encapsulation.
- Each VLAN is assigned a gateway IP address on its corresponding sub-interface, allowing devices in that VLAN to send traffic to other VLANs through the router.

```
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        unassigned      YES unset  up                    up
FastEthernet0/0.10     192.168.10.1    YES manual up                    up
FastEthernet0/0.20     192.168.20.1    YES manual up                    up
FastEthernet0/0.30     192.168.30.1    YES manual up                    up
FastEthernet0/0.40     192.168.40.1    YES manual up                    up
```

Figure 6: IP Interface

8. SECURITY IMPLEMENTATION (ACL)

- Using an **Access Control List (ACL)**, rules were configured on the router to control traffic between VLANs.
- A deny rule was applied to block the HR VLAN (VLAN 10) from accessing the Server VLAN (VLAN 40), preventing unauthorized access.
- Permit rules were added to allow IT (VLAN 20) and Admin (VLAN 30) to access the Server VLAN, ensuring only authorized departments can communicate with servers.

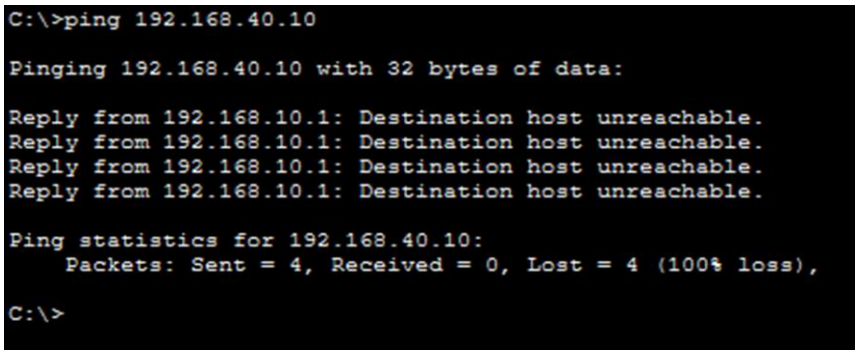
```
access-list 100 permit ip 192.168.30.0 0.0.0.255 any
access-list 100 permit ip 192.168.20.0 0.0.0.255 host 192.168.40.10
access-list 100 deny ip 192.168.10.0 0.0.0.255 host 192.168.40.10
access-list 100 permit ip any any
```

Figure 7: ACL Configuration

Explanation:

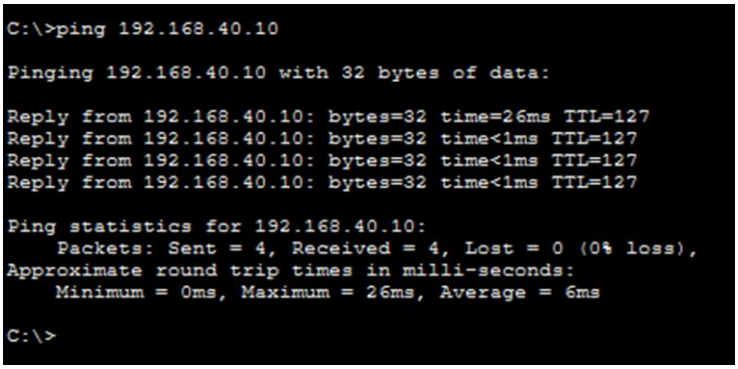
- Allows Admin network full access
- Allows IT to access only the server
- Denies HR access to the server
- Allows all other traffic.

Test Result



```
C:\>ping 192.168.40.10

Pinging 192.168.40.10 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 192.168.40.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Figure 8: Ping request from HR to Server



```
C:\>ping 192.168.40.10

Pinging 192.168.40.10 with 32 bytes of data:

Reply from 192.168.40.10: bytes=32 time=26ms TTL=127
Reply from 192.168.40.10: bytes=32 time<1ms TTL=127
Reply from 192.168.40.10: bytes=32 time<1ms TTL=127
Reply from 192.168.40.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.40.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 26ms, Average = 6ms

C:\>
```

Figure 9:Ping request from IT to Server

Testing & Validation

The ACL was tested using connectivity checks:

- **HR → Server (192.168.40.10)** → Failed
- **IT → Server** → Successful
- **Admin → Server** → Successful

This confirms that the ACL is functioning as intended.

9. SERVER SETUP

A dedicated server was configured in a separate VLAN to simulate a real-world application environment. This server provides services to authorized users while remaining isolated from unauthorized access through network security controls.

9.1 VLAN Configuration

The server was placed in:

- **VLAN 40 (Server Network)**

This separation ensures that server resources are isolated from user networks (HR, IT, Admin), improving security and traffic management.

9.2 IP Configuration

The server was assigned a **static IP address** to ensure consistent accessibility:

- **IP Address:** 192.168.40.10
- **Subnet Mask:** 255.255.255.0
- **Default Gateway:** 192.168.40.1

9.3 Service Configuration

The **HTTP service** was enabled on the server to simulate a web application:

- HTTP service turned **ON**
- Allows users to access the server via web browser
- Example access :[http://192.168.40.10]

9.4 Security Consideration

- The server is protected using **ACL rules**
- Only authorized VLANs (IT and Admin) can a
- Unauthorized users (HR) are restricted

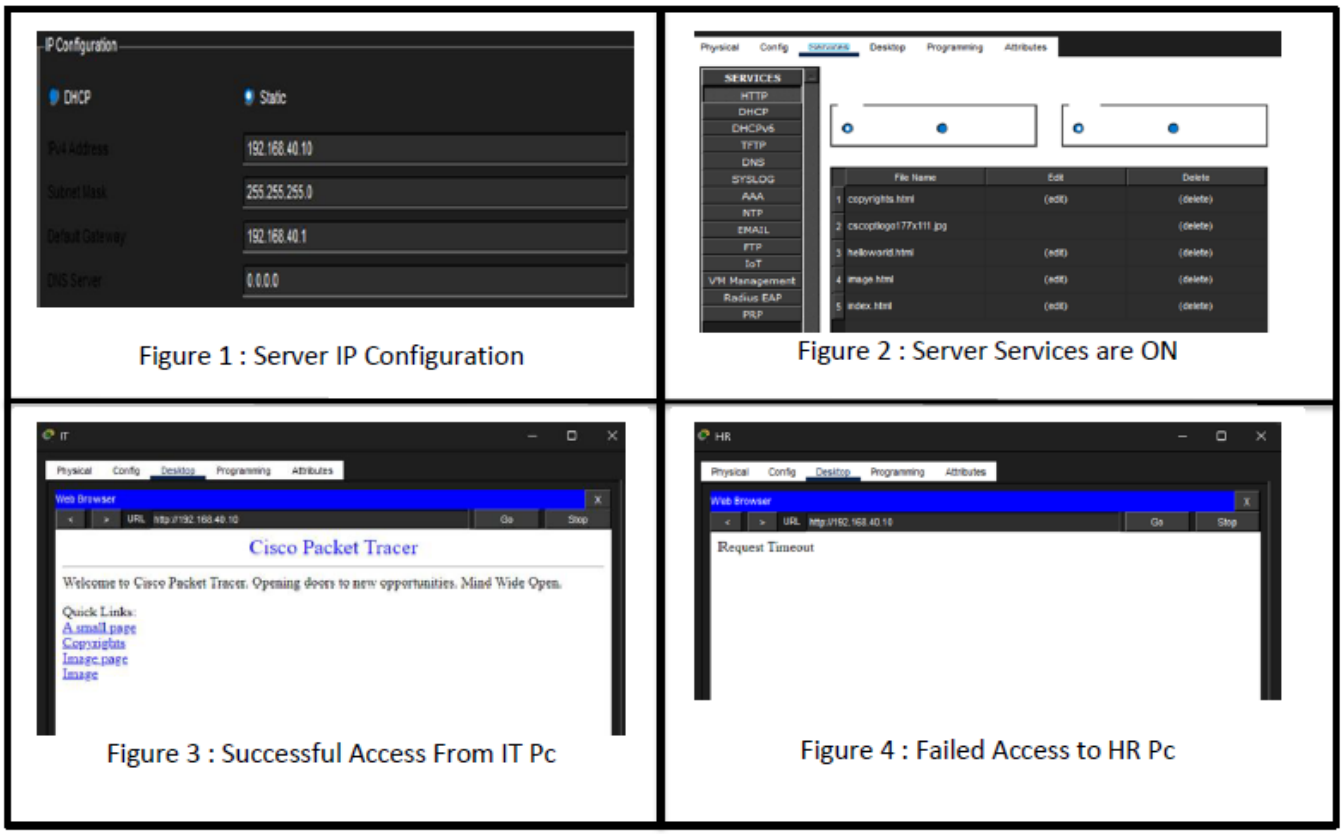


Figure 10 : HTTP Access To Server

10. PORT SECURITY

Port security was implemented on the switch to prevent unauthorized devices from connecting to the network. This feature enhances Layer 2 security by restricting access based on MAC addresses.

10.1 Configuration Details

Port security was enabled on access ports connected to end devices (HR, IT, and Admin PCs).

Key configurations:

- **Port Security Enabled:**
  Ensures that only trusted devices can connect to the switch port.
- **Sticky MAC Learning:**
  The switch automatically learns and stores the MAC address of the first connected device. This MAC address is then bound to the port.
- **Maximum MAC Addresses:**
  Limited to **1 device per port**, preventing multiple devices from connecting.
- **Violation Mode – Shutdown:**
  If an unauthorized device is detected (different MAC address), the port is automatically disabled.

Working Principle

1. When a valid device connects, its MAC address is learned and stored.
2. If another device with a different MAC address connects to the same port:
    - A **security violation** occurs
    - The port enters **secure-shutdown state**
    - Network access is blocked

11. ATTACK SIMULATION

A rogue device attack was simulated to test the effectiveness of port security. This type of attack represents an unauthorized user attempting to gain network access by connecting a device to an active switch port.

11.1 Attack Scenario

- An attacker device was connected to a **secured switch port**
- The port was already configured with **port security and sticky MAC learning**
- The legitimate device's MAC address was previously learned and stored.

11.2 Detection & Response

- The switch detected a **MAC address mismatch**
- A **security violation** was triggered
- The port automatically entered **secure-shutdown state**
- Network access was immediately blocked

```
Switch#show port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
              (Count)       (Count)        (Count)
--------------------------------------------------------------------------
      Fa1/1       1            1              0          Shutdown
      Fa2/1       1            1              0          Shutdown
      Fa3/1       1            1              0          Shutdown
--------------------------------------------------------------------------
Switch#
```

Figure 1 : Overall Port Security

```
Switch#show port-security interface fa1/1
Port Security               : Enabled
Port Status                 : Secure-up
Violation Mode              : Shutdown
Aging Time                  : 0 mins
Aging Type                  : Absolute
SecureStatic Address Aging  : Disabled
Maximum MAC Addresses       : 1
Total MAC Addresses         : 1
Configured MAC Addresses    : 0
Sticky MAC Addresses        : 1
Last Source Address:Vlan    : 0090.2B38.1E6D:10
Security Violation Count    : 0
```

Figure 2 : Before Attack

```
Switch#show port-security interface fa1/1
Port Security               : Enabled
Port Status                 : Secure-shutdown
Violation Mode              : Shutdown
Aging Time                  : 0 mins
Aging Type                  : Absolute
SecureStatic Address Aging  : Disabled
Maximum MAC Addresses       : 1
Total MAC Addresses         : 1
Configured MAC Addresses    : 0
Sticky MAC Addresses        : 1
Last Source Address:Vlan    : 0003.E401.2C69:10
Security Violation Count    : 1
```

Figure 3 : After Attack

```
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface fa1/1
Switch(config-if)#shutdown

%LINK-5-CHANGED: Interface FastEthernet1/1, changed state to administratively down
Switch(config-if)#no shutdown

Switch(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/1, changed state to up
```

Figure 4 : Reset Interface

Figure 11: Port Security Test

12. DHCP ATTACK

The DHCP starvation attack is a type of **Denial of Service (DoS)** attack where an attacker attempts to exhaust all available IP addresses from the DHCP server. This prevents legitimate users from obtaining IP addresses and accessing the network.

12.1 Attack Concept

In a normal scenario:

- A client sends a DHCP request
- The server assigns an available IP address

In a DHCP starvation attack:

- Multiple fake clients (or attacker systems) send repeated DHCP requests
- The DHCP server assigns IP addresses to each request
- The **IP address pool gets exhausted**

12.2 Simulation in Lab

In the lab environment, this concept was demonstrated by:

- Connecting multiple client devices to the same VLAN
- Configuring all devices to obtain IP addresses via DHCP
- Each device requesting an IP address from the DHCP server

12.3 Observation

- DHCP server assigns IPs to multiple clients
- The available IP pool decreases rapidly
- New devices fail to obtain IP addresses

```
Router>enable
Router#show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
192.168.10.2     0090.2B38.1E6D           --                     Automatic
192.168.20.2     00D0.FF83.6BB2           --                     Automatic
192.168.30.2     0004.9A98.E5A0           --                     Automatic
Router#
```

Figure 12 : DHCP IP allocation

12.4 Prevention Techniques

To protect against DHCP starvation attacks:

- **Port Security:** Limits number of devices per port
- **DHCP Snooping (Advanced):** Filters unauthorized DHCP requests
- **MAC Address Limiting:** Restricts multiple MAC addresses on a port

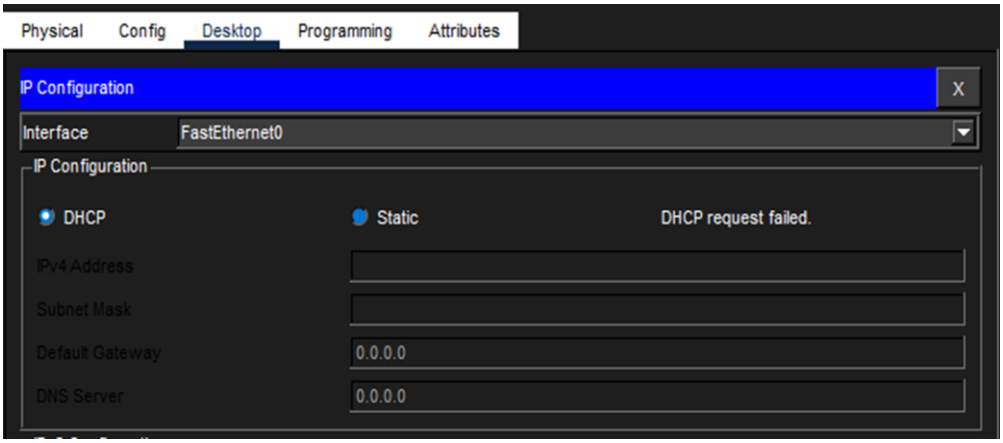


Figure 13: DHCP Request Failed for new PC

13. RESULTS & TESTING

Test Case	Expected Result	Actual Result	Status
DHCP Configuration	Devices receive IP automatically	IP assigned successfully	Passed
Inter-VLAN Communication	VLANs communicate via router	Communication successful	Passed
ACL Implementation	HR blocked from Server	Access denied	Passed
Server Access (IT/Admin)	Allowed access to server	Access successful	Passed
Port Security	Unauthorized device blocked	Port shutdown triggered	Passed
Attack Simulation	Attack detected and mitigated	Attack blocked	Passed

All configurations were validated using ping tests, DHCP allocation checks, and access verification.

14. LIMITATIONS

- Simulated environment using Cisco Packet Tracer
- No real hardware implementation
- Limited attack scenarios (only basic attacks tested)
- Basic security mechanisms (ACL, Port Security only)
- Not designed for large-scale networks

15. FUTURE ENHANCEMENTS

- Implement Intrusion Detection and Prevention Systems (IDS/IPS)
- Integrate firewall for advanced network security
- Enable real-time network monitoring
- Use tools like Wireshark and Snort for traffic analysis
- Implement DHCP Snooping and Dynamic ARP Inspection
- Expand the network for enterprise-level scalability

16. CONCLUSION

This project demonstrates a secure network architecture using VLAN segmentation, routing, and access control mechanisms. Security features such as ACL and port security effectively prevented unauthorized access. Attack simulations validated the robustness of the implemented defenses, highlighting the importance of layered network security.

17. REFERENCES

- Cisco Networking Documentation
- RFC 2131 – DHCP Protocol
- Network Security Fundamentals Guides

This activity was performed in a controlled lab environment for educational purposes only.